

Strings

A string is simply a sequence of characters (text data).

Strings are immutable.

Python allows 4 ways to define strings, they are

1. By using Single Quotes

Use for simple Text

Ex: `s1 = 'AVD Group'`

2. By using Double Quotes

Use double quotes When your text contains single quotes inside

Ex: `s1 = "It's Good Day"`

3. By using Triple single Quotes

Use for multi-line Strings

Ex: `data = '''101,siva,50000
102,kiran,60000'''`
`print(data)`

4. By using Triple Double Quotes

Use when you want to create document strings

Ex:

`log = """ERROR: Disk full
Time: 10:30PM """`
`print(log)`

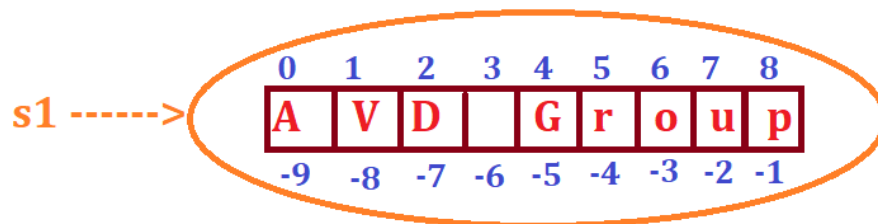
→ We can access string data in two ways

1. By using forward index which starts with 0

2. By using Backward index which starts with -1 from last element onwards

Ex:

`s1 = 'AVD Group'`



--> We can display String data in 4 ways

1. We can display string directly by using variable name

2. We can display string character by character by using for loop

3. We can display specific character by using index number

4. We can display part of the data by using slicing

Display all characters at a time by using variable name

```
s1 = 'AVD Group'  
print(s1)
```

Display specific characters by using index number

```
s1 = 'AVD Group'
print(s1[0]) # A
print(s1[4]) # G
```

Display part of string by using slicing

```
s1 = 'AVD Group'
print(s1[0:3]) # AVD
print(s1[:]) # AVD Group
print(s1[::-1]) # puorG DVA (Reverse String)
```

Display all character one by one by using for loop

```
s1 = 'AVD Group'
for i in s1:
    print(i,end='\t')
```

→ Write a program to read a string then print only vowels

```
s1 = 'avd group'
for i in s1:
    if i in 'aeiou':
        print(i,end='')

```

→ Write a program to read string then print only Lowercase characters

```
s1 = 'AvD GRoup'
for i in s1:
    if i>='a' and i<='z':
        print(i,end='\t')
```

→ Write a program to read string then print only digits

```
s1 = 'A2vD GRo5u2p'
for i in s1:
    if i>='0' and i<='9':
        print(i,end='\t')
```

→ Write a program to read string then print only symbols

```
s1 = 'A2v#D GR@o5u$2p'
for i in s1:
    if not ('a'<=i<='z' or 'A'<=i<='Z' or
            '0'<=i<='9' or i==' '):
        print(i,end='\t')
```

Methods

1. **upper()** : Used to convert string into upper case

2. **lower()** : Used to convert string into lower case

```
s1 = 'AVD Group'
s2 = s1.lower()
s3 = s1.upper()
print(s2)
print(s3)
```

3. If you want to compare values then we have 3 options

1. By using **==** operator

2. By using **startswith()** method

3. By using **endswith()** method

Note:

If you want to compare complete string then use **==**

If you want to compare starting part of string then use **startswith()** method

If you want to compare ending part of string then use **endswith()** method

```
Ex2: code = '1122'
      ht = '1122CSE001'
      if(ht.startswith(code)):
          print('Belongs to our college')
```

```
Ex3: emailid = 'Venkat@ibm.com'
      if(emailid.endswith('ibm.com')):
          print('Belongs to ibm company')
```

5. **index()** : It is used to get index number of a given character

6. **count()** : It is used to count no. of occurrence of a given character

```
s1 = 'AVD Group'
ch = input('Enter any character : ')
i = s1.index(ch)
print(i)
```

```
s1 = 'AVD Group'
ch = input('Enter any character : ')
i = s1.count(ch)
print(i)
```

7. replace() : It is used to replace existing character/string with new character/string

```
s1='core java and adv java both are imp for java'
s2 = s1.replace('j','b')
s3 = s1.replace('java','Python')
print(s2)
print(s3)
```

8. split() : It is used to divide given string into multiple parts based on given character, incase user doesn't specify any character then it will split based on space.

Ex:1 `s1 = 'Core Python and Adv Python Both are imp for Python'`
 `s2 = s1.split()`
 `print(s2)`

Ex2: `s1 = 'Vishnu@tcs.com'`
 `s2=s1.split('@')`
 `print(s2)`
 `print('Emp Name is : ',s2[0])`

9.strip() : It is used to remove all spaces before and after the string

```
s1 = '  AVD Group  '
s2 = s1.lstrip()
s3 = s1.rstrip()
s4 = s1.strip()
print(s2, ' ', s3, ' ', s4)
```

10. **.join()**: Used to concatenate string

```
words = ["Data", "Engineer"]  
print(" ".join(words))
```

11. **format()**: used to insert values into a string dynamically

```
name = "Vikram"  
status = "SUCCESS"  
log = "User {} login {}".format(name, status)  
print(log)
```

12. **isupper()**: used to check given string in uppercase or not

13. **islower()**: used to check given string in lowercase or not

14. **isdigit()**: used to check given string is digit or not

15. **isalnum()**: used to check string may alphabet or digit

```
s1=input('Enter any name : ')  
for i in s1:  
    if i.islower():  
        print(i,end='\t')
```

Note: It will display lower case characters only

len() function

➔ **len()** function is used to find length of a given string

```
s1='avd group'  
print(len(s1))
```